Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The area bounded by $\mathrm{y}-1=|\mathrm{x}|, \mathrm{y}=0$ and $|\mathrm{x}|=\frac{1}{2}$ will be:

(A) $\frac{3}{4}$

(B) $\frac{3}{2}$

(C) $\frac{5}{4}$

(D) None of these

Q.2

Find the area bounded by the curve $x=2-y-y^2$ and y -axis.

(A) $-\frac{9}{2}$

(B) $\frac{9}{2}$

(C) 9

(D) -9

Q.3

Area between the curve $y=\cos^2 x, x$ -axis and ordinates $\mathrm{x}=0$ and $\mathrm{x}=\pi$ in the interval $(0, \pi)$ is

(A) $\frac{2\pi}{3}$

(B) 2π

(C) π

(D) $\frac{\pi}{2}$

Q.4

The coordinates of the point $\mathrm{P}(\mathrm{x}, \mathrm{y})$ of $\mathrm{y}=\mathrm{e}^{-|\mathrm{x}|}$ so that the area formed by the coordinate axes and the tangent at P is greatest, are

(A) $(1,1/\mathrm{e})$

(B) $(1,-1/\mathrm{e})$

(C) $(1,\mathrm{e})$

(D) $(-1,\mathrm{e})$

@IITJEE_Advanced

Q.5

The area under the curve $y = |\cos x - \sin x|, 0 \leq x \leq \frac{\pi}{2}$, and above x -axis is :

(A) $2\sqrt{2}$

(B) $2\sqrt{2} - 2$

(C) $2\sqrt{2} + 2$

(D) 0

Q.6

The area of the figure bounded by the curves $y = |x - 1|$ and $y = 3 - |x|$ is

(A) 1

(B) 3

(C) 2

(D) 4

Q.7

If the area enclosed by $\mathrm{y}^2 = 4\mathrm{ax}$ and line $\mathrm{y} = \mathrm{ax}$ is $\frac{1}{3}$.sq.unit, then the area enclosed by $y = 4x$ with same parabola is $\frac{4}{3}$ sq unit. Find the roots of parabola.

(A) -4 and 2

(B) 2 and 4

(C) -2 and -4

(D) 8 and -8

Q.8

Let $\mathrm{f(x)}$ be a continuous function such that the area bounded by the curve $\mathrm{y} = \mathrm{f(x)}, \mathrm{x}$ -axis and the lines $\mathrm{x} = 0$ and $\mathrm{x} = \mathrm{a}$ is $\frac{\mathrm{a}^2}{2} + \frac{\mathrm{a}}{2}\sin \mathrm{a} + \frac{\pi}{2}\cos \mathrm{a}$, then
$f\left(\frac{\pi}{2}\right) =$

(A) 1

(B) $\frac{1}{2}$

(C) $\frac{1}{3}$

(D) None of these

Q.9

The area of the smaller segment cut off from the circle $x^2 + y^2 = 9$ by $x = 1$ is

@IITJEE_Advanced

(A) $\frac{1}{2}\left(9\sec^{-1}3-\sqrt{8}\right)$ sq unit

(B) $\left(9\sec^{-1}3-\sqrt{8}\right)$ sq unit

(C) $\left(\sqrt{8}-9\sec^{-1}3\right)$ sq unit

(D) None of the above

Q.10

What is the area of the parabola $x^2=y$ bounded by the line $y=1$?

(A) $\frac{1}{3}$ square unit

(B) $\frac{2}{3}$ square unit

(C) $\frac{4}{3}$ square units

(D) 2 square units

Q.11

What is the area of the region enclosed by $\mathrm{y}=2\,|x|$ and $y=4$

(A) 2 square unit

(B) 4 square unit

(C) 8 square unit

(D) 16 square unit

Q.12

The area of the region (in sq. units), in the first quadrant bounded by the parabola $y=9x^2$ and the lines $x=0, y=1$ and $y=4$, is:

(A) $7/9$

(B) $14/3$

(C) $7/3$

(D) $14/9$

Q.13

Let the area bounded by the x-axis, curve $y=1+\frac{8}{x^2}$ and the ordinates $x=2$ and $x=4$ is "A" sq. unit and if the ordinate $x=$ a divides the area into two equal parts, then the correct statement among the following is

(A) $A=3$ and $a=2$

(B) $A=2$ and $a=1$

(C) $A=4\sqrt{2}$ and $a=\sqrt{2}$

@IITJEE_Advanced

(D) $A = 4$ and $a = 2\sqrt{2}$

Q.14

The value of $a(a > 0)$ for which the area bounded by the curves $y = \frac{x}{6} + \frac{1}{x^2}, y = 0, x = a$ and $x = 2a$ has the least value is

(A) 2

(B) $\sqrt{2}$

(C) $2^{1/3}$

(D) 1

Q.15

If $f(x) = a + bx + cx^2$, where $c > 0$ and $b^2 - 4ac < 0$ then the area enclosed by the coordinate axes, the line $x = 2$ and the curve $y = f(x)$ is given by

(A) $\frac{1}{3}\{4f(1) + f(2)\}$

(B) $\frac{1}{2}\{f(0) + 4f(1) + f(2)\}$

(C) $\frac{1}{2}\{f(0) + 4f(1)\}$

(D) $\frac{1}{3}\{f(0) + 4f(1) + f(2)\}$

Q.16

The area enclosed between the curve $\mathrm{y} = \log_{\mathrm{e}}(\mathrm{x} + \mathrm{e})$ and the coordinate axes is

(A) 1

(B) 2

(C) 3

(D) 4

Q.17

The area bounded by the curve $y = f(x), y = x$ and the lines $x = 1, x = t$ is $\left(t + \sqrt{1 + t^2}\right) - \sqrt{2} - 1$ squnit, for all $t > 1$. If $f(x)$ satisfying $f(x) > x$ for all $x > 1$, then $f(x)$ is equal to

(A) $x + 1 + \frac{x}{\sqrt{1+x^2}}$

(B) $x + \frac{x}{\sqrt{1+x^2}}$

(C) $1 + \frac{x}{\sqrt{1+x^2}}$

(D) $\frac{x}{\sqrt{1+x^2}}$

@IITJEE_Advanced

Q.18

The value of $c+2$ for which the area of the figure bounded by the curve $y=8x^2-x^5$, the straight lines $x=1$ and $x=c$ and x -axis is equal to $\frac{16}{3}$, is

(A) 1

(B) 3

(C) -1

(D) 4

Q.19

What is the area bounded by $y=\tan x, y=0$ and $x=\frac{\pi}{4}$?

(A) $\ln 2$ square units

(B) $\frac{\ln 2}{2}$ square units

(C) $2(\ln 2)$ square units

(D) None of these

Q.20

What is the area of the portion of the curve $y=\sin x$, lying between $x=0, y=0$ and $x=2\pi$?

(A) 1 square unit

(B) 2 square units

(C) 4 square units

(D) 8 square units

Answers & Solutions

Q.1 Answer:

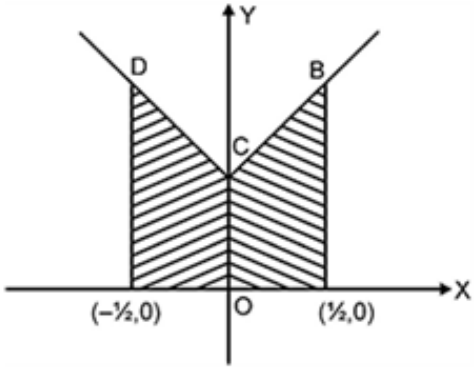
$\frac{5}{4}$

Solution:

The given lines are, $\mathrm{y}-1=\mathrm{x}, \mathrm{x} \geq 0; \quad \mathrm{y}-1=-\mathrm{x}, \mathrm{x}<0$
$\mathrm{y}=0; \quad \mathrm{x}=-\frac{1}{2}, \mathrm{x}<0; \quad \mathrm{x}=\frac{1}{2}, \mathrm{x} \geq 0$

@IITJEE_Advanced

so that the area bounded is as shown in the figure.



Required area

$= 2\int_0^{1/2}(1+x)\mathrm{dx} = 2\left(\mathrm{x}+\frac{\mathrm{x}^2}{2}\right)_0^{1/2} = 2\left(\frac{1}{2}+\frac{1}{8}\right) = \frac{5}{4}$

Q.2 Answer:

$\frac{9}{2}$

Solution:

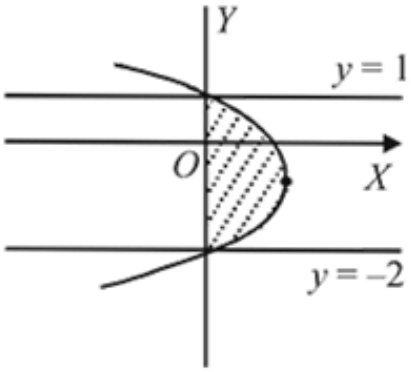
Put $2-y-y^2=0$

$\Rightarrow y=1,-2$

This means, the curve intersects the y **-axis at** $y=1$ **and** $y=-2$.

Hence required area $=\int_{-2}^{1} x dy$

$=\int_{-2}^{1}\left(2-y-y^2\right) dy$



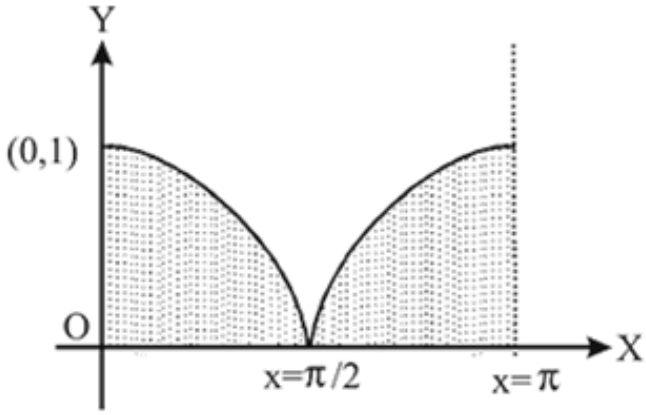
$=\left[2y-\frac{y^2}{2}-\frac{y^3}{3}\right]_{-2}^{1}=\frac{9}{2}$ **sq. units**

Q.3 Answer:

$\frac{\pi}{2}$

Solution:

@IITJEE_Advanced



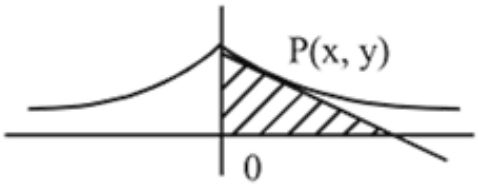
Required area $= \int_0^{\pi} \cos^2 x dx$
$= \int_0^{\pi/2} \cos^2 \mathrm{x} d\mathrm{x} + \int_{\pi/2}^{\pi} \cos^2 \mathrm{x} d\mathrm{x}$
$= \frac{1}{2} \times \frac{\pi}{2} + \frac{1}{2} \int_{\pi/2}^{\pi} (1 + \cos 2\mathrm{x}) d\mathrm{x}$
$= \frac{\pi}{4} + \frac{1}{2} \left[\mathrm{x} + \frac{\sin 2\mathrm{x}}{2}\right]_{\pi/2}^{\pi} = \frac{\pi}{4} + \frac{1}{2} \left[(\pi - \frac{\pi}{2})\right]$
$= \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$

Q.4 Answer:

$(1, 1/\mathrm{e})$

Solution:

Graph of $\mathrm{y} = \mathrm{e}^{-|\mathrm{x}|}$ **is symmetrical about** y **-axis, so we consider** $x \geq 0$, **then** $y = e^{-x} \Rightarrow \frac{dy}{dx} = -e^x$



Tangent at P **is** $\mathrm{Y} - \mathrm{y} = -\mathrm{e}^{-\mathrm{x}}(\mathrm{X} - \mathrm{x})$
Its x **-intercept** $= x + ye^{-x}$ **and** y **-intercept** $= y + xe^{-x}$
So area $A = \frac{1}{2} \left(x + ye^{-x}\right) \left(y + xe^{-x}\right)$
$= \frac{1}{2}(x+1)(1+x)e^{-x} \quad [\because y = e^{-x}]$
$= \frac{1}{2}(x+1)^2 \mathrm{e}^{-\mathrm{x}}$
$\frac{\mathrm{dA}}{\mathrm{dx}} = (\mathrm{x}+1)\mathrm{e}^{-\mathrm{x}} - \frac{1}{2}(\mathrm{x}+1)^2 \mathrm{e}^{-\mathrm{x}} = \frac{1}{2}(\mathrm{x}+1)(1-\mathrm{x})\mathrm{e}^{-\mathrm{x}}$
where A **is maximum if** $\mathrm{x} = 1$ **So** P **is** $\left(1, \mathrm{e}^{-1}\right)$ **Due to symmetry, there is another point** $\left(-1, \mathrm{e}^{-1}\right)$

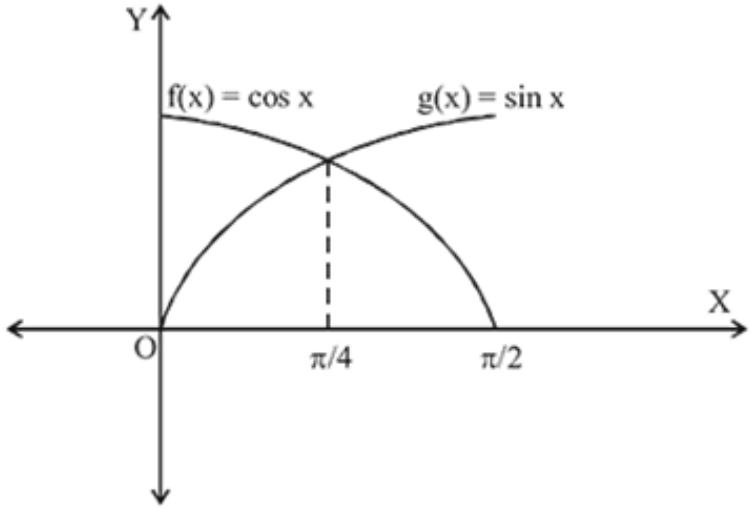
Q.5 Answer:

$2\sqrt{2} - 2$

Solution:

@IITJEE_Advanced

$y = |\cos x - \sin x|$



Required area $= 2\int_0^{\pi/4}(\cos x - \sin x)dx$
$= 2[\sin x + \cos x]_0^{\pi/4} = 2\left[\frac{2}{\sqrt{2}} - 1\right] = (2\sqrt{2} - 2)$ **sq. units**

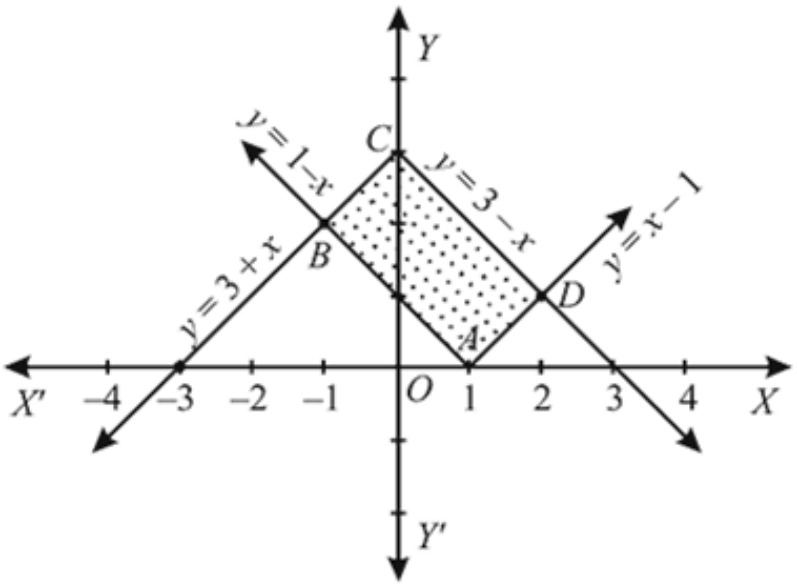
Q.6 Answer:

4

Solution:

The given curves are :

$$\left.\begin{matrix} y = x - 1, x \geq 1 \\ y = 1 - x, x \leq 1 \end{matrix}\right\} \text{ and } \left.\begin{matrix} y = 3 - x, & x > 0 \\ y = 3 + x, & x < 0 \end{matrix}\right\}$$



The required area = area of $ABCD$
$= \int_{-1}^{0}(3 + x)dx + \int_0^2(3 - x)dx$
$- \int_{-1}^{1}(1 - x)dx - \int_1^2(x - 1)dx$
$= \frac{5}{2} + 4 - 2 - \frac{1}{2} = 4$ **sq. units.**

@IITJEE_Advanced

Q.7 Answer:

-4 and 2

Solution:

$y = \int_0^{4/a}(a \cdot x - \sqrt{4a \cdot x})dx$
$\frac{1}{3} = \int_0^{4/a} axdx - \int_0^{4/a} \sqrt{4ax}dx$
$\frac{1}{3} = \left[\frac{ax^2}{2}\right]_0^{4/a} - 2\left[\frac{(4ax)^{3/2}}{3}\right]_0^{4/a}$
$\frac{1}{3} = \frac{16a}{a^2} - \frac{2}{3}\left[4a\left(\frac{4}{a}\right)^{3/2}\right], a = 8$

Putting the value of a in $x^2 + 2x - a = 0$, we get its roots i.e., -4 and 2.

Q.8 Answer:

$\frac{1}{2}$

Solution:

We have,

$\int_0^a f(x)dx = \frac{a^2}{2} + \frac{a}{2}\sin a + \frac{\pi}{2}\cos a$

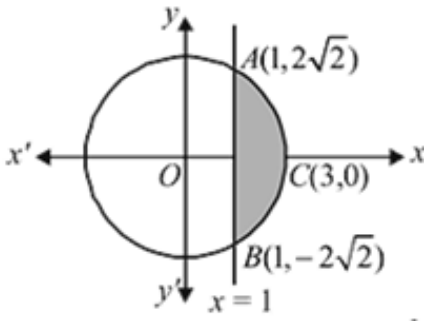
Differentiating w.r.t. a, we get $\mathrm{f(a)} = \mathrm{a} + \frac{1}{2}(\sin \mathrm{a} + \mathrm{a}\cos \mathrm{a}) - \frac{\pi}{2}\sin \mathrm{a}$

Put $\mathrm{a} = \frac{\pi}{2}; \mathrm{f}\left(\frac{\pi}{2}\right) = \frac{\pi}{2} + \frac{1}{2} - \frac{\pi}{2} = \frac{1}{2}$

Q.9 Answer:

$\left(9\sec^{-1}3 - \sqrt{8}\right)$ sq unit

Solution:



Given, equation of the circle is $x^2 + y^2 = 9$.

$\therefore$ Area of the smaller segment cut off from the circle $x^2 + y^2 = 9$ by $x = 1$, is given by

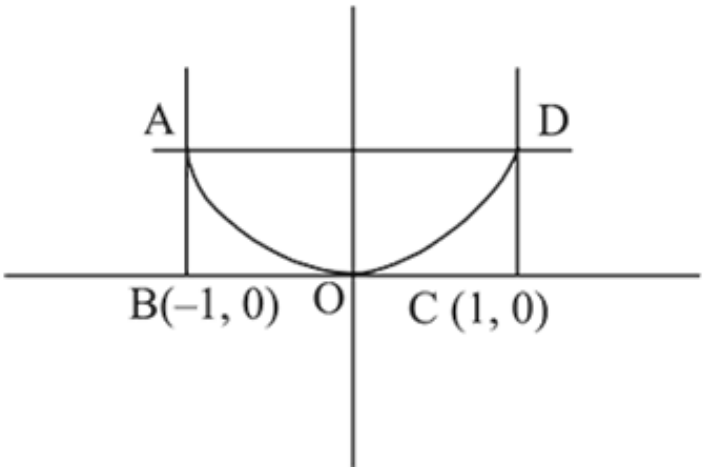
$A = 2\int_1^3 \sqrt{9 - x^2}dx = 2 \cdot \frac{1}{2}\left[x\sqrt{9 - x^2} + 9\sin^{-1}\frac{x}{3}\right]_1^3$
$= \left[3 \cdot \sqrt{9 - 9} + 9\sin^{-1}\left(\frac{3}{3}\right) - 1 \cdot \sqrt{9 - 1} - 9\sin^{-1}\left(\frac{1}{3}\right)\right]$
$= \left[9\sec^{-1}(3) - \sqrt{8}\right]$ **sq. unit.**

Q.10 Answer:

$\frac{4}{3}$ square units

Solution:

@IITJEE_Advanced



Area of $ABCD = 2 \times 1 = 2$ **sq. units**

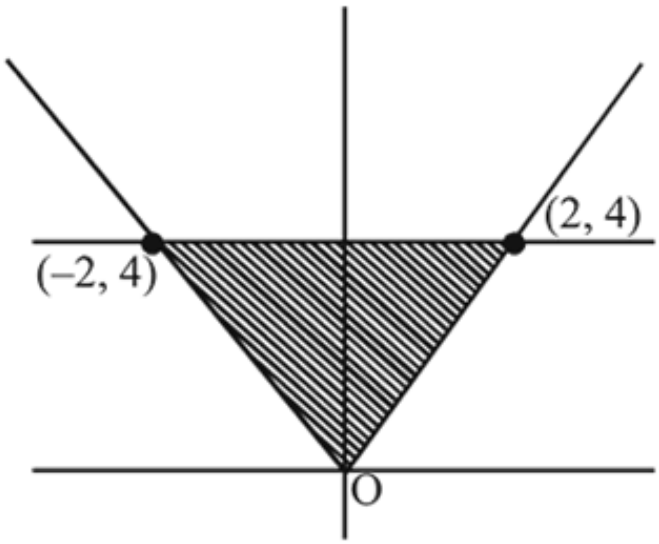
Area of $BAODC = \int_{-1}^{1} x^2 dx = \frac{x^3}{3}\Big|_{-1}^{1} = \frac{2}{3}$ **sq. units**

Required area $= 2 - \frac{2}{3} = \frac{4}{3}$ **sq. units**

Q.11 Answer:

8 square unit

Solution:



Required Area $= 2\int_{y=0}^{4} \frac{y}{2} dy = \frac{y^2}{2}\Big|_{0}^{4} = \frac{16}{2} = 8$ **sq. unit.**

or Area $= 2 \times 4 = 8$ **sq. unit.**

Q.12 Answer:

$14/9$

Solution:

@IITJEE_Advanced

$$\text{Required area } = \int_{y=1}^{4} \sqrt{\frac{y}{9}}dy = \frac{1}{3}\int_{y=1}^{4} y^{1/2}dy$$

$$= \frac{1}{3} \times \frac{2}{3}\left|\left(y^{3/2}\right)\right|_1^4 = \frac{2}{9}\left[\left(4^{1/2}\right)^3 - \left(1^{1/2}\right)^3\right]$$

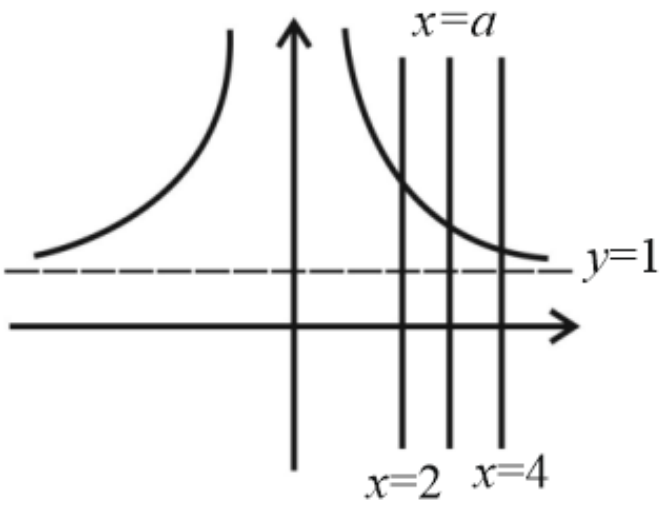
$$= \frac{14}{9} \text{ sq. units.}$$

Q.13 Answer:

$A = 4$ and $a = 2\sqrt{2}$

Solution:

$y = 1 + \frac{8}{x^2}$



Required area $A = \int_2^4 ydx = \int_2^4 1 + \frac{8}{x^2}dx$

$= x - \frac{8}{x}\Big|_2^4 = 4$

If $x = a$ **bisects the area then we have** $\int_2^a 1 + \frac{8}{x^2} = \frac{4}{2}$ ... 1

Now $\int_2^a 1 + \frac{8}{x^2}dx = x - \frac{8}{x}\Big|_2^a = a - \frac{8}{a} - 2 + 4$

Now $a - \frac{8}{a} + 2 = 2$ **(by equation (1))**

$\Rightarrow a - \frac{8}{a} = 0 \Rightarrow a^2 = 8$

$\Rightarrow a = \pm 2\sqrt{2}$

Since $2 < a < 4$

$\therefore a = 2\sqrt{2}$

Q.14 Answer:

1

Solution:

@IITJEE_Advanced

$$f(a) = \int_a^{2a} \left(\frac{x}{6} + \frac{1}{x^2}\right) dx = \left(\frac{x^2}{12} - \frac{1}{x}\right)_a^{2a}$$
$$= \left(\frac{4a^2}{12} - \frac{1}{2a} - \frac{a^2}{12} + \frac{1}{a}\right) = \frac{a^2}{4} + \frac{1}{2a}$$

Let $f'(a) = \frac{2a}{4} - \frac{1}{2a^2} = 0$

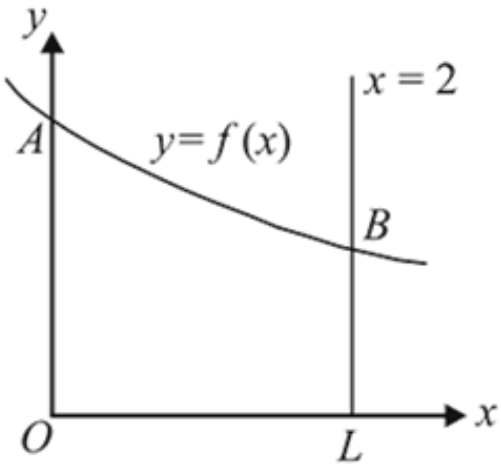
$\Rightarrow a = 1$ **which is a point of minima.**

Q.15 Answer:

$\frac{1}{3}\{f(0) + 4f(1) + f(2)\}$

Solution:

Area of $OABL = \int_0^2 y dx$



$= \int_0^2 \left(a + bx + cx^2\right) dx = 2a + 2b + \frac{8}{3}c$

$= \frac{1}{3}[6a + 6b + 8c] \ldots$ **(i)**

But, $f(x) = a + bx + cx^2; f(0) = a, f(1) = a + b + c$

$f(2) = a + 2b + 4c \Rightarrow \frac{1}{3}\{f(0) + 4f(1) + f(2)\}$

$= \frac{1}{3}\{a + 4(a + b + c) + (a + 2b + 4c)\} = \frac{1}{3}\{6a + 6b + 8c\}$

Q.16 Answer:

1

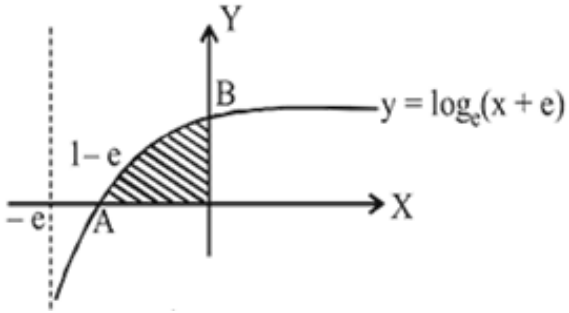
Solution:

Required area (OAB)

$= \int_{1-e}^0 \ln(x + e) dx$

@IITJEE_Advanced

$= \left[x \ln(x+e) - \int \frac{1}{x+e} x dx\right]_0^1 = 1$



Q.17 Answer:

$x + 1 + \frac{x}{\sqrt{1+x^2}}$

Solution:

It is given that, $f(x) > x$**, for all** $x > 1$**. So, area bounded by** $y = f(x), y = x$ **and the lines** $x = 1, x =$**t**
is given by $\int_1^t \{f(x) - x\} dx$
But this area is given equal to $\left(t + \sqrt{1+t^2} - \sqrt{2} - 1\right)$ **sq unit. Therefore,** $\int_1^t \{f(x) - x\} dx = t + \sqrt{1+t^2} - \sqrt{2} - 1$**, for all** $t > 1$
On differentiating both sides w.r.t. t**, we get**
$f(t) - t = 1 + \frac{t}{\sqrt{1+t^2}}$ **for all** $t > 1$
$\Rightarrow f(t) = t + 1 + \frac{t}{\sqrt{1+t^2}}$ **for all** $t > 1$
Hence, $f(x) = x + 1 + \frac{x}{\sqrt{1+x^2}}$ **for all** $x > 1$

Q.18 Answer:

1

Solution:

For $c < 1, \int_c^1 \left(8x^2 - x^5\right) dx = \frac{16}{3}$
$\Rightarrow \frac{8}{3} - \frac{1}{6} - \frac{8c^3}{3} + \frac{c^6}{6} = \frac{16}{3}$
$\Rightarrow c^3 \left[-\frac{8}{3} + \frac{c^3}{6}\right] = \frac{17}{6}.$
Again, for $c \geq 1$**, none of the values of** c **satisfy the required condition that** $\int_1^c \left(8x^2 - x^5\right) dx = \frac{16}{3} \Rightarrow c + 2 = 1$

Q.19 Answer:

$\frac{\ln 2}{2}$ square units

Solution:

Required area $= \int_0^{\frac{\pi}{4}} \tan x dx$
$= \ln|\sec x|_0^{\frac{\pi}{4}} = \ln \sqrt{2} = \frac{\ln 2}{2}$

Q.20 Answer:

@IITJEE_Advanced

4 square units

Solution:

The graph of $y = \sin x$ is shown in image.

$A = \int_0^{2\pi} (\sin x)$

$A = \int_0^{\pi} (\sin x) + \int_{\pi}^{2\pi} (\sin x)$

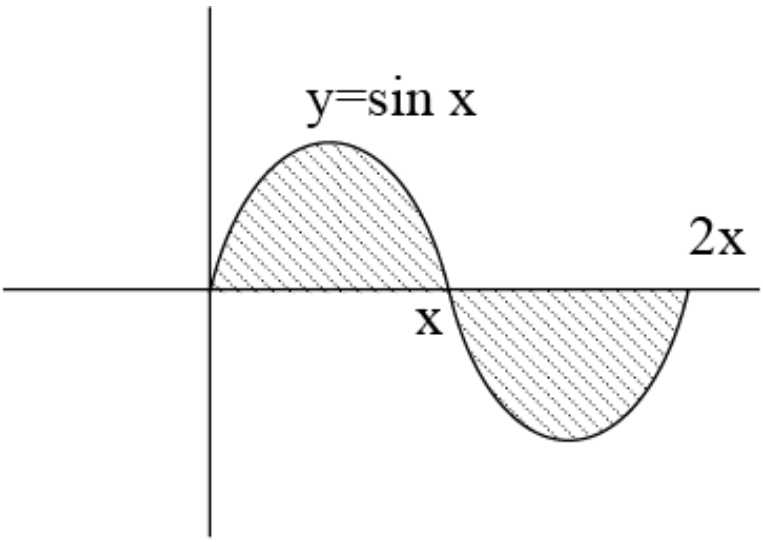
Since, $\left|\int_0^{\pi} (\sin x)\right| = \left|\int_0^{2\pi} (\sin x)\right|$

Hence, $\mathrm{A} = 2\left(\int_0^{\pi} (\sin \mathrm{x})\right)$

$\mathrm{A} = -2[\cos \mathrm{x}]\Big|_{\pi}^{0}$

$\mathrm{A} = 2 \times 2$

$\mathrm{A} = 4$ sq. unit



@IITJEE_Advanced